Reihaneh Malekian

Ph.D. Candidate in Statistics, Columbia University
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Education

Columbia University, New York, NY — Expected May 2027
Ph.D. in Statistics
Advisor: Sumit Mukherjee

Sharif University of Technology, Tehran, Iran
M.Sc. in Pure Mathematics

Sharif University of Technology, Tehran, Iran
B.Sc. in Mathematics and Applications

Research Interests

Probability; large deviations; random graphs and hypergraphs; Gibbs measures and statistical mechanics; high-dimensional statistics; robust inference for networks and dependent data.

Publications and Preprints

- **Malekian, R.**, Bhattacharya, S., Deb, N., and Mukherjee, S. **LDP for Tensor Forms**. 2026
  arXiv:2609.02682.
- **Malekian, R.**, Xing, H., and Maleki, A. **Is Speckle Noise More Challenging to Mitigate Than Additive Noise?**, *IEEE Transactions on Information Theory*, 71(8):6353–6372. 2025
- **Malekian, R.**, and Ramdas, A. **Matrix Concentration: Order versus Anti-order**. 2024
  arXiv:2408.05998.

Manuscripts in Preparation

- "Fluctuations in Mean-Field Potts Models," with Sumit Mukherjee. Manuscript nearing completion; draft available upon request.
- "Statistical Inference for Online Least-Squares SGD under Stationary Dependence," with Promit Ghosal.
- "Empirical Bayes Inference with Gaussian Scale Mixture Priors," with Taehyun Kim.

Selected Research Presentations

- Upcoming invited talk, MIT Probability Seminar, Massachusetts Institute of Technology, November 2026.
- Talk, Yale Student Theory Lunch, Yale University, October 2026.
- Invited talk, Probability Seminar, Stanford, September 2026.
- Invited talk, Probability Seminar, UC Berkeley, September 2026.
- Contributed talk, Joint Statistical Meetings, Boston, August 2026.
- Talk, ICERM Hot Topics Workshop, Brown University, August 2026.
- Poster, International Congress of Mathematicians, Philadelphia, July 2026.
- Poster, Peter Hall and Bob Shumway Memorial Conference on Dependent Data Analysis in the Modern Era, UC Davis, May 2026.
- Invited talk, Finger Lakes Probability Seminar, Syracuse University, April 2026.
- Seminar talk, Columbia University Student Seminar, February 2026.
- Talk, Progress in Discrete and Continuous Probability Conference, Union College, Schenectady, NY, July 2025.
- Poster, INFORMS Applied Probability Society Conference, Atlanta, July 2025.
- Poster, Bernoulli–IMS World Congress in Probability and Statistics, Bochum, Germany, August 2024.
- Talk, Minghui Yu Memorial Conference, Columbia University, April 2024, 2025, and 2026.

Honors and Awards

- Teaching Observation Fellow, Columbia University Center for Teaching and Learning, 2026–2027.
- Early Career Travel Award, American Statistical Association (ASA), Joint Statistical Meetings, 2026.
- Graduate Research Fellowship, Columbia University, 2022–2027.
- Ranked second in overall GPA in both the B.Sc. and M.Sc. cohorts, Sharif University of Technology.
- Recipient of undergraduate and graduate scholarships from Iran's National Elites Foundation.
- Admitted to the graduate program without an entrance examination as an exceptionally talented B.Sc. student, Sharif University of Technology.
- Placed among the top 0.05% of participants in Iran's nationwide university entrance examination.

Selected Workshops and Summer Schools

ICERM Hot Topics Workshop
Synergies between Geometry, Probability, and Computation in High Dimensions, Brown University — August 2026
SLMath Summer Graduate School
Concentration Inequalities and Localization Techniques in High-Dimensional Probability and Geometry — July 2023

Teaching Experience

Teaching Assistant, Columbia University — 2022–Present

- Introduction to Statistics; Statistical Inference; Linear Regression Models; Introduction to Statistical Reasoning; Introduction to Probability and Statistics; Probability.

Teaching Assistant, Sharif University of Technology

- Probability and Applications; Probability Theory; Mathematical Writing; Algebra I; Analysis I.

Research and Academic Mentoring

Directed Reading Program Mentor, Columbia University — Fall 2026
Mentoring an undergraduate student in probability through an individualized semester-long reading project.
Bachelor's Project Co-supervisor, EPFL — 2021
Co-supervised Clémence Marcaillou's bachelor's project in combinatorial discrepancy theory through weekly research meetings.

Selected Expository Writing

- "Good Numbers." *Pargar*, Year 2, no. 4, pp. 17–23, Spring 2014. [In Persian]
- "The Turán Number and Probabilistic Combinatorics," by Alan J. Aw. Persian translation and adaptation of the article published in *The American Mathematical Monthly*, 119(6):510–513, 2012. *Pargar*, Year 2, no. 8, pp. 9–13, Spring 2015. [In Persian]

Outreach and Academic Service

Graduate Student Mentor, GROW 2026, Columbia University — 2026
First-Year PhD Peer Mentor, Department of Statistics, Columbia University — 2023, 2024, 2026
Organizing Team Member and Program Moderator, Maryam Mirzakhani Celebration, Sharif University of Technology, Tehran, Iran — May 2019
Student Organizer, Undergraduate Mathematics Summer School, IASBS, Zanjan, Iran — 2019
Student Organizer, Mathematics Summer School for High School Students, Borujen, Iran — 2015

Skills

Programming: Python, R.
Languages: Persian (Farsi; native), English (proficient), French (intermediate, B1).